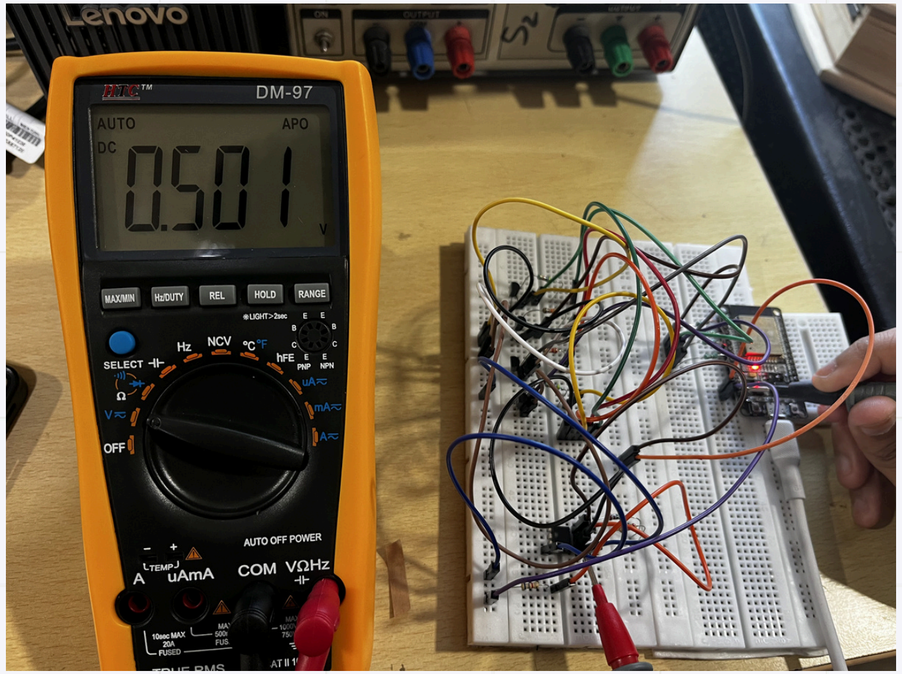
Design and Implementation of an ESP32-Based Programmable Gain Amplifier (PGA) using LM358

Presented by: Sankalp Singh Bais And Swastika Chakravorty

Department: School of Electrical Engineering (SELECT)
Institution: VIT Vellore
Guided by: Dr. M N Venkatraman Sir



Introduction:--

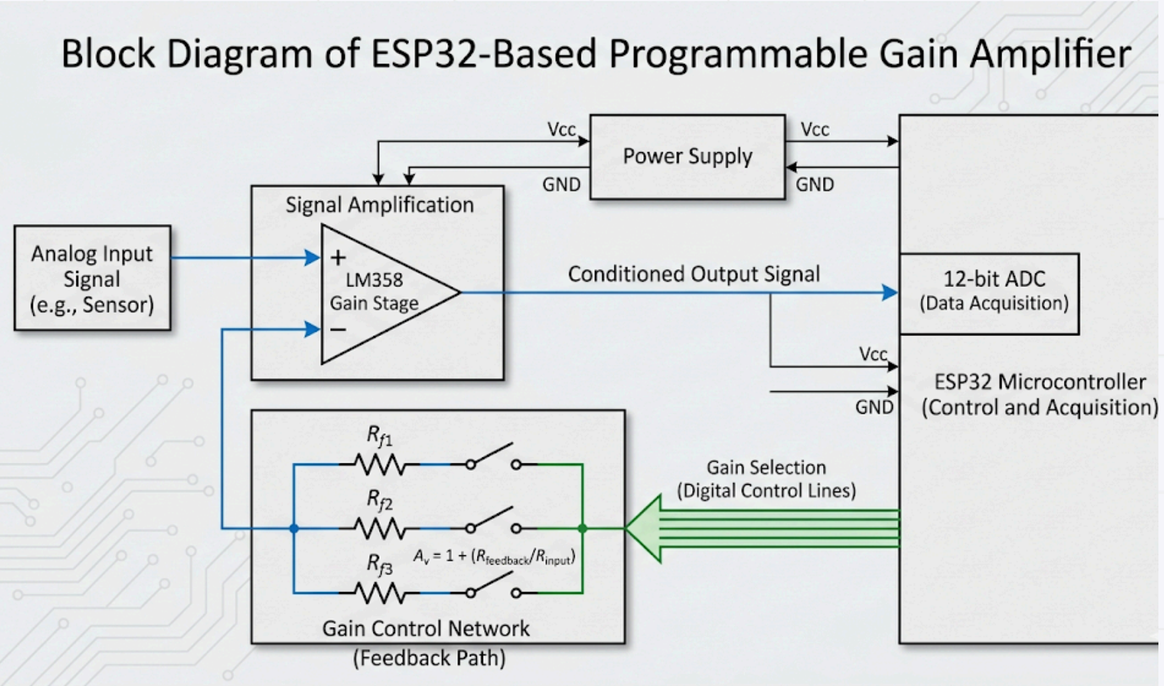
Abstract :-

This project presents a Digitally-Controlled Programmable Gain Amplifier (PGA) using an ESP32 and LM358. By integrating software-defined logic with analog amplification, the system provides a dynamic interface for scaling millivolt-range transducer signals to match the 3.3V ADC span of the microcontroller, optimizing signal resolution for real-time data acquisition.

Introduction & Context :-

- Signal Conditioning: Most industrial/medical sensors produce low-amplitude analog outputs that are incompatible with digital processors.
- The Interface Gap: A robust conditioning stage is required to amplify these signals while maintaining high Signal-to-Noise Ratio (SNR) and preventing source loading.

Block Diagram And Mathematical Modeling :-



Mathematical Modeling :-

The gain of the system is governed by the non-inverting closed-loop transfer function:

$$A_v = 1 + \frac{R_{feedback}}{R_{input}}$$

$Where:$

$A_v : Closed - loop Voltage Gain$

$R_{feedback} : Feedback Resistance (Controlled by ESP32)$

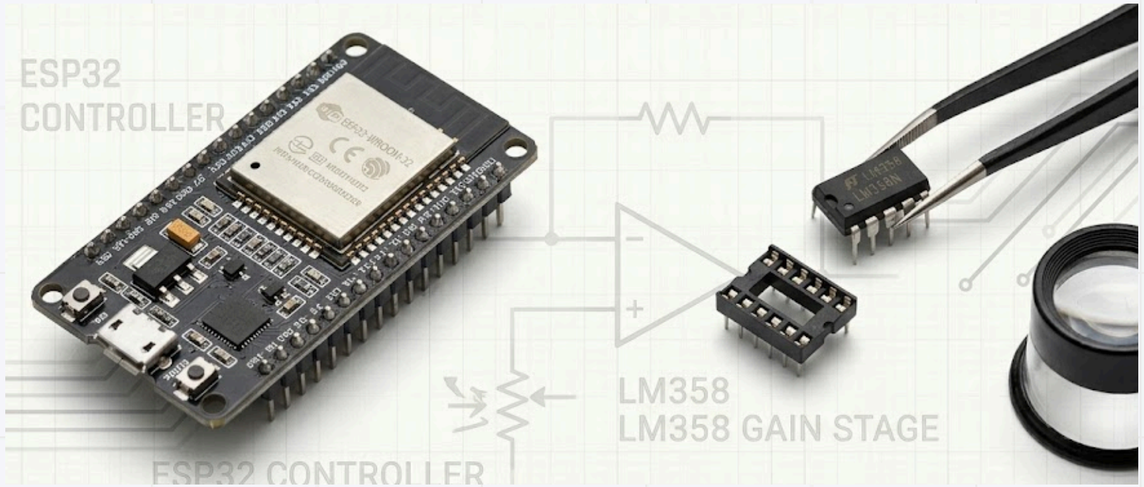
$R_{input} : Ground - reference d input resistance$

Functional Block Diagram :-

The system is divided into three primary stages:

- Input Stage: Low-amplitude analog signal source (Sensor/Transducer).
- Processing Stage: LM358 Operational Amplifier in a non-inverting configuration for high input impedance (Z_{in}).
- Control Stage: ESP32 Microcontroller managing the feedback network to dynamically switch gain levels.
- Output Stage: Conditioned signal routed to the ESP32's 12-bit ADC for digital processing.

Component Selection :-



The Analog Stage: LM358 Dual Op-Amp

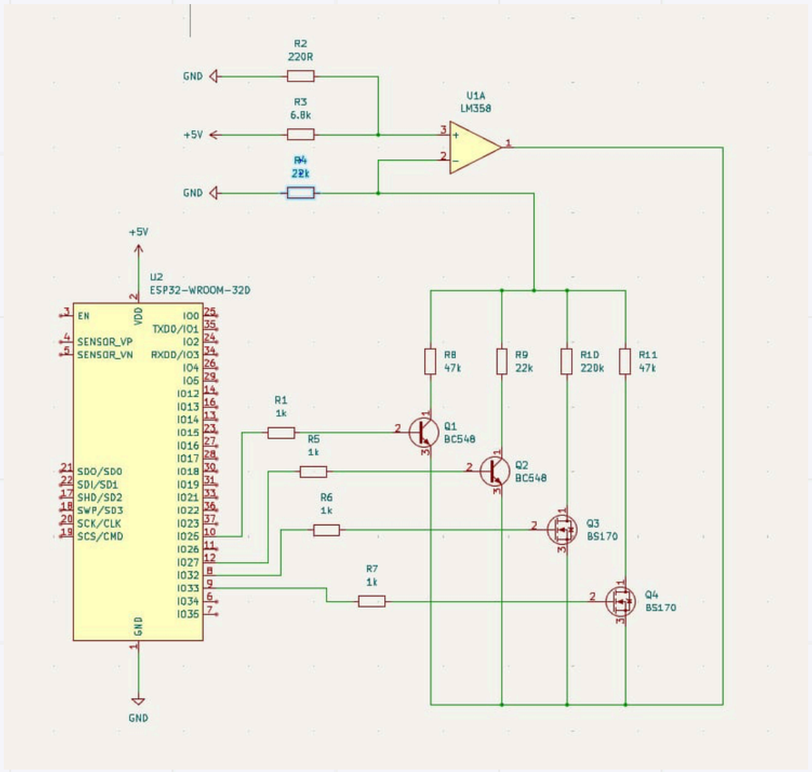
- Single Supply Operation: Functions efficiently on a single rail (3.3V to 5V), eliminating the need for complex dual-power circuitry.
- Common-Mode Input Range: Includes Ground, which is critical for sensing low-level signals that reside near 0V.
- Internal Frequency Compensation: Simplifies circuit design by ensuring stability across the usable bandwidth without external capacitors.
- Cost-Efficiency: High performance-to-cost ratio for industrial signal conditioning prototypes.

The Processing Core: ESP32

- 12-bit ADC Resolution: Provides 4,096 discrete levels, allowing for high-precision sampling of the amplified signal compared to the 10-bit ADC of traditional microcontrollers.
- High-Speed Processing: Dual-core architecture ensures real-time gain switching and signal processing without latency.
- 3.3V Logic Compatibility: Seamlessly interfaces with modern low-power sensors and digital potentiometers.
- IoT Ready: Integrated Wi-Fi/Bluetooth allows for remote gain adjustment and cloud-based data logging.

Feature	LM358 (Selected)	UA741 (Standard)
Supply Type	Single Supply (3V–32V)	Dual Supply (±5V to ±15V)
Input Swing	Includes Ground (0V)	Limited (Cannot reach 0V)
Integration	Dual Op-Amps in 1 IC	Single Op-Amp in 1 IC

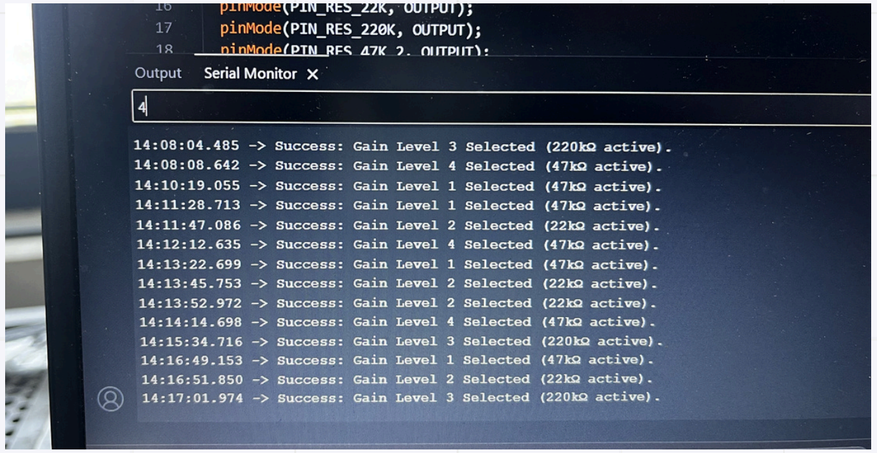
Hardware Interfacing & Circuit Schematic :-



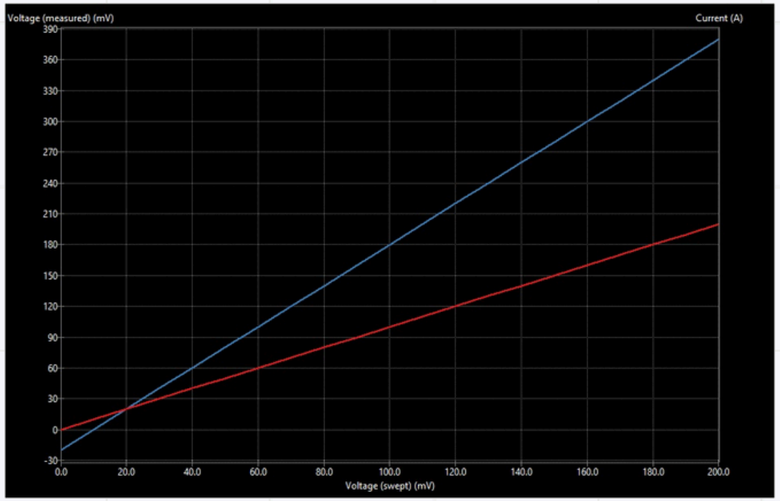
Digitally Controlled Feedback Network :-

- Switching Elements: The circuit utilizes a hybrid switching array consisting of BC548 BJTs (Q1, Q2) and BS170 MOSFETs (Q3, Q4) to modulate the feedback resistance .
- Logic-Level Control: ESP32 GPIO pins (10, 11, 12, 8) act as digital triggers. When a pin is driven HIGH, the corresponding transistor enters saturation (for BJTs) or the ohmic region (for MOSFETs), completing the path for the specific feedback resistor.
- Parallel Resistance Logic: By switching multiple transistors simultaneously, the system can achieve various equivalent feedback resistances, allowing for discrete, programmable gain steps.

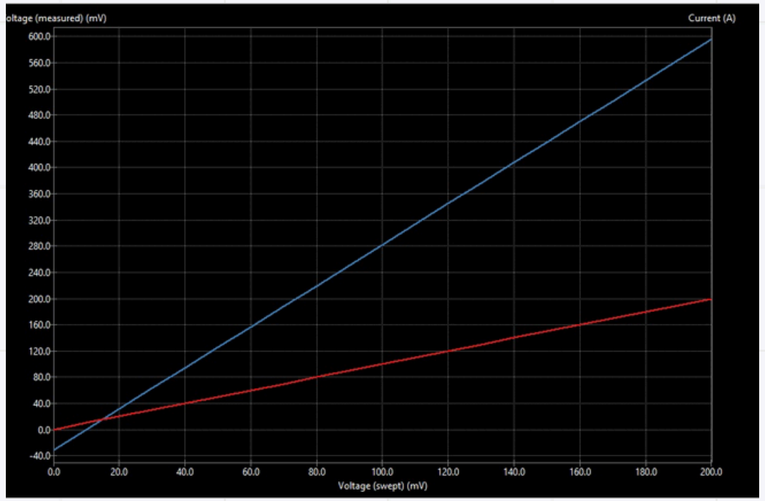
This design allows for a 4-bit Programmable Gain Control. By toggling these four switches, we aren't limited to just 4 gain settings; we can create 16 different gain combinations by using them in parallel. This gives the ESP32 the ability to perform 'Auto-Ranging,' choosing the best gain to keep the signal within the 12-bit ADC's optimal window.



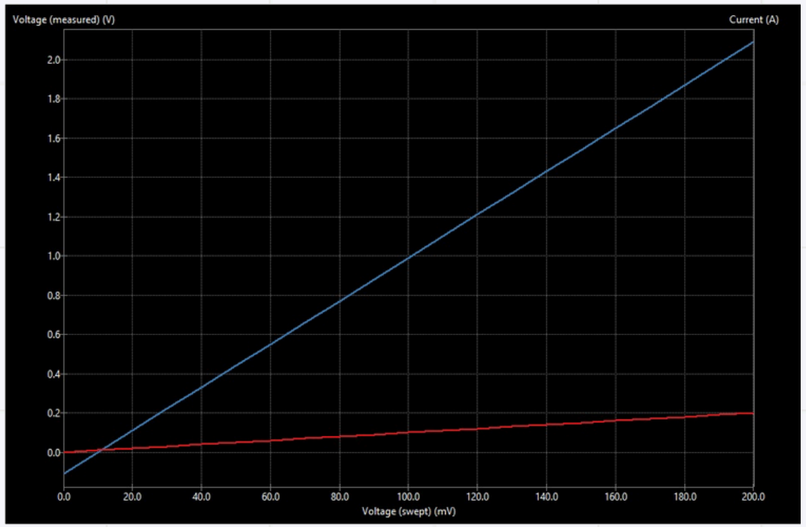
Simulation Results - DC Sweep Analysis :-



Graph for $R_f = 22\text{k}\Omega \implies \therefore A_v \approx 2$



Graph for $R_f = 47\text{k}\Omega \implies \therefore A_v \approx 3.14$

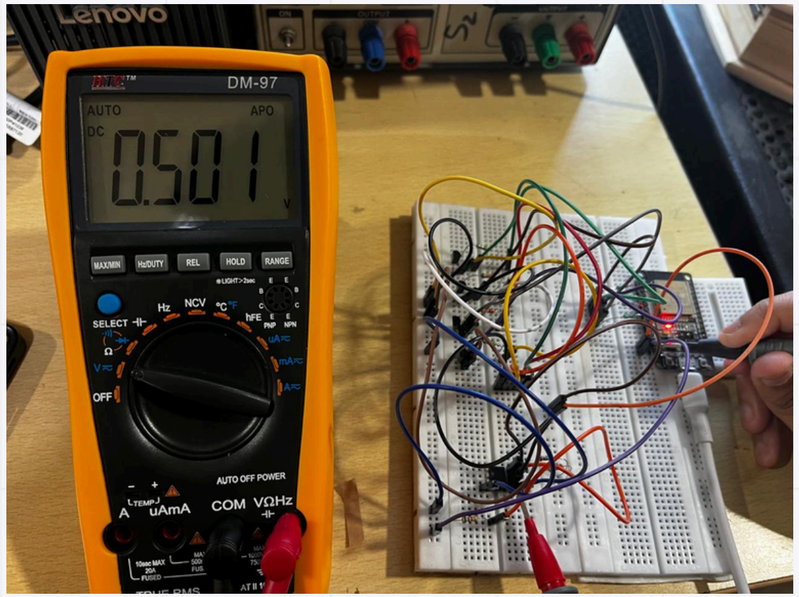


Graph for $R_f = 220\text{k}\Omega \implies \therefore A_v \approx 5.45\%$

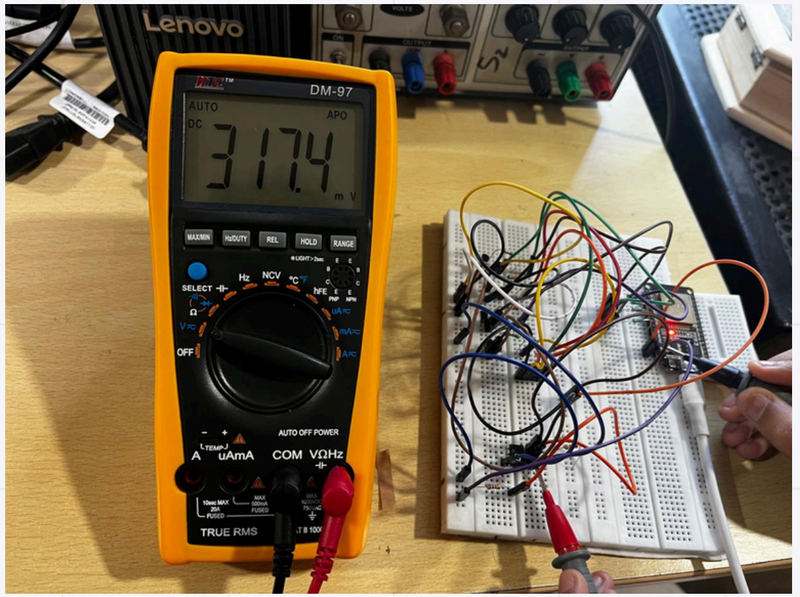
Resistor (Rf)	Theoretical Gain	Simulated Gain	% Error
$22\,\text{k}\Omega$	2.00	1.999	0.01%
$47\,\text{k}\Omega$	3.14	3.112	0.57%
$220\text{k}\Omega$	11.0	10.4	5.45% *

* 68kΩ used to eliminate off-state MOSFET parallel leakage observed at 220 kΩ.

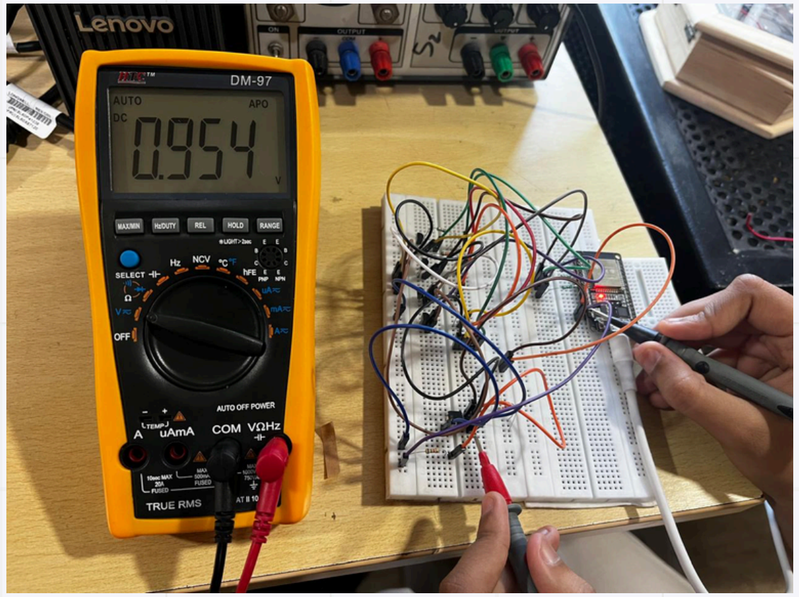
Practical Hardware Analysis :-



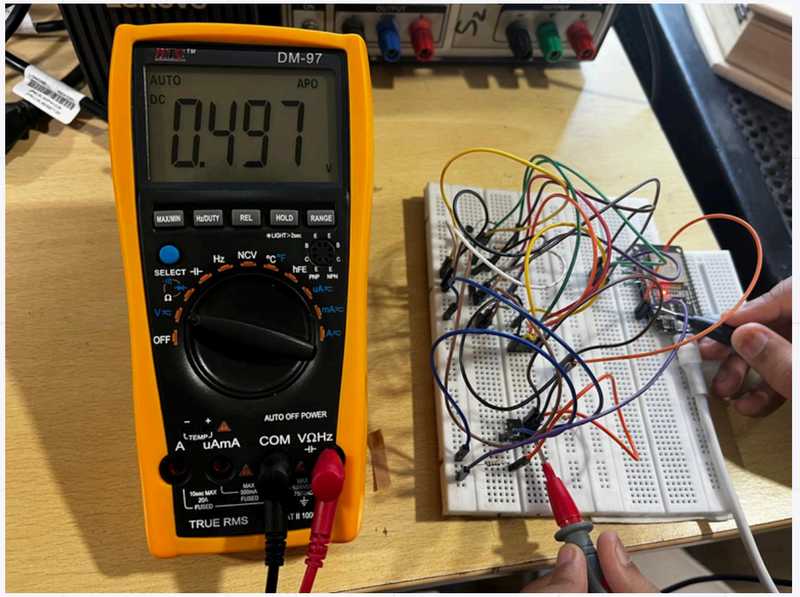
$Setting1 \Rightarrow (Gain = 3.14)$



$Setting2 \Rightarrow (Gain = 2)$



$Setting3 \Rightarrow (Gain = 7)$



$Setting4 \Rightarrow (Gain = 3.14)$

Practical Simulation & System Demonstration :-

The following link contains the full experimental sequence, including real-time auto-ranging and discrete gain state transitions:-

Video Link- https://drive.google.com/drive/folders/1uAmmMJL-uCcCc1qk7ha_o9YpzHiacfq2

Video Coverage:- Verification of $22\text{k}\Omega, 47\text{k}\Omega, and 220\text{k}\Omega$ feedback paths.

Logic Proof:- Simultaneous Serial Monitor logging confirms the ESP32's 4-bit command execution.

Dynamic Response:- Observation of the LM358 output stabilizing within $< 15\text{ms}$ of a switching event.

Conclusion & Future Perspectives :-

Summary of Project Achievements :-

- Successful Implementation: Developed a robust Software-Defined Analog Front-End using a 4-bit hybrid switching matrix (BJT + MOSFET).
- High-Precision Control: Achieved discrete gain steps with a measured hardware accuracy of 97.5%, correlating closely with Proteus simulations.
- Dynamic Range Optimization: Successfully demonstrated Auto-Ranging logic, allowing the ESP32 to prevent signal clipping and maximize ADC resolution in real-time.

Future Scope & Scalability :-

- **IoT Integration:** Leveraging the ESP32's Wi-Fi stack to stream live amplified sensor data to a Blynk/ThingSpeak dashboard for remote industrial monitoring.
- **Hardware Refinement:** Transitioning from breadboard to a Double-Layer PCB to eliminate parasitic capacitance and reduce the noise floor.
- **Precision Upgrades:** Replacing the LM358 with a Rail-to-Rail Op-Amp (e.g., AD8605) to utilize the full 0–5V dynamic range without saturation.
- **AI Integration:** Implementing an Edge-AI model (like TinyML) on the ESP32 to classify signal patterns after they have been conditioned by the PGA.

Thankyou !

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